

Scope of Probabilistic Methods.

PROBABILISTIC METHODS is a modified name for the subject, formerly STATISTICAL DECISION SUPPORT, and even earlier PROBABILISTIC METHODS AND STATISTICS.

The subject consists of three parts:

Probability

– a branch of mathematics that studies the laws governing random phenomena, meaning those whose real detailed course or outcome cannot be determined before their occurrence.

It is the mathematical foundation of statistics, playing a crucial role in situations requiring the analysis of large data sets.

The theory first found its application in the *analysis of games of chance in the 17th century*.

Mathematical statistics

– a branch of mathematics that deals with statistical inference methods, which involve drawing conclusions about the entire population based on the results obtained from a sample.

Statistical inference involves the estimation and verification of statistical hypotheses.

Statistical decision theory

- the basic tool is the decision function, which assigns decisions to the results of observations or the results of appropriate statistical experiments;
in such a way that the benefits associated with making an correct decision are as maximally as possible or losses are as minimal as possible.

I. Mathematical statistics

1. DESCRIPTION OF THE ANALYZED ELEMENTS.

The analyzed elements are characterized by a variety of features/properties.

Variants of a feature are its possible values.

The following types of features/properties are distinguished:

- **Measurable (quantitative) features**

- a feature whose variants are *numbers (expressed in units)*.

Example: age (year of birth), height (in cm), weight (in kg), quantity (expressed in integer number)

- **Immeasurable (qualitative) features**

- a feature whose variants are not numbers.

Example: colour of hairs, beauty, emotions, shadow of white, loudness.

Depending on the relationships between the feature's variants, we speak of **different scales**.

We distinguish four scales: *nominal, ordinal, interval, and ratio*:

- **Nominal scale**

- allows us to determine whether two variants are **equal or different**.

- **Ordinal scale**

- allows us to determine for any two variants whether they are equal, and if not, which one is less than the other, i.e., an **ordering relation is introduced** in the set of variants.

- **Interval scale**

- a scale whose variants are numbers with units, where subtraction of variants is specified (in the sense that the difference between variants makes sense given the meanings of the variants), i.e., it is possible to determine by how many units one variant is greater or smaller than another.

- **Ratio scale**

- a scale whose variants are numbers with units, where subtraction and division of variants are specified.

Example:

Age and the number of overdue exams are measurable (quantitative) features, while the features: gender, marital status, eye color, and statistics grade are *immeasurable (qualitative) features*.

→ Age and the number of overdue exams are features on **a ratio scale**.

→ The statistics grade is a feature on **an ordinal scale**.

→ Gender and marital status are expressed on **a nominal scale**.

Scales can be arranged as follows: Ratio scale, Interval scale, Ordinal scale, Nominal scale, in the sense that any scale appearing in the above sequence is also each of the scales that follows it.

Another way to express this is to say that the nominal scale is the lowest level, followed by the ordinal scale, interval scale, and the highest level, the ratio scale. It is possible to transform a higher-level scale into a lower-level scale.

The type of scale determines the statistical methods used.

2. FEATURES AS RANDOM VARIABLES.

It is convenient to study **measurable features because their variants are real numbers**, so statistical analysis can be performed using mathematical concepts.

However, the means for studying immeasurable features are limited.

To avoid this problem and to proceed uniformly when examining measurable and immeasurable features, we **define a random variable X as follows**:

- *If the feature is measurable*, then each element of the population is assigned with a value (variant) of that element's feature;

- *If the feature is immeasurable*, then elements with the same feature value are assigned with the same real number, and population elements with different feature values are assigned with different real numbers.

We assume that **the random variable X** thus defined is a *model of the population feature*, and its distribution (X distribution) is the distribution of that feature in the population.

Features transformed in this way are always measurable.

However, when analyzing them, it is always necessary to consider what features they were before the transformation, in order to use concepts that make sense for the meaning of the features under consideration.

[Frequency Table Calculator - Create Frequency Distribution \(statistics\)](#)

3. Frequency Table Calculator

A frequency

is the number of times a data value occurs.

For example, if ten students score 90 in statistics, then score 90 has a frequency of 10.

A frequency is a count of the occurrences of values within a data-set.

Cumulative frequency

is used to determine the number of observations below a particular value in a data set.

The cumulative frequency is calculated by adding each frequency from a frequency distribution table to the sum of its predecessors.

The last value will always be equal to the total for all data.

A relative frequency

is a frequency divided by a count of all values.

Relative frequencies can be written as *fractions, percents, or decimals*.

Cumulative relative frequency

is the accumulation of the previous relative frequencies.

The last value will always be equal to 1.



How to enter grouped data?

Grouped data are formed by aggregating individual data into groups so that a frequency distribution of these groups serves as a convenient means of summarizing or analyzing the data.

group	frequency
10-20	5
20-30	10
30-40	15

This grouped data can be entered as:

10-20 20-30 30-40

f: 5 10 15

How to enter data as a cumulative frequency table?

Similar to a frequency table, but instead, f: write cf: in the second line. For example:

$n = 80$; $k = 8$

cumulative quantity: 10 20 30 40 50 60 70 80

cumulative frequency: 5 13 20 32 60 80 90 100

Task : Draw table of frequency

Group	Frequency
10 -20	5
20-30	8
30-40	7

The cumulative frequency is calculated by adding each frequency from a frequency distribution table to the sum of its predecessors. The last value will always equal the total for all observations since the calculator will have already added all frequencies to the previous total.

Key words:

mean, standard deviation, modus, median, percentile, decile, quartile, range

Practice problems from statistics:

- Problem 1: Calculate 45 percentile for given data.



Given the following data 11 15 24 33 10 35 23 25 40

What is P45?

- Problem 2. Employees - statistical



The company has 18 employees aged 26-52. The age groups of the employees are: 3 employees aged 52 years, 2 aged 32 years, 1 ... 26 years old, 5 ... 36 years old, 4 ... 45 years old, and 3 ... 50 years old. Determine the median.

- Decile

Find the 5.5th decile of the data: 62, 60, 37, 57, 55, 59, 57, 50, 49, 61

- A batsman

A batsman scored the following number of runs in seven innings 35,30,45,65,39,20,40. Find the mean, median, and range.

- The median 2

Here is a list of numbers: 9.9, 5.9, 3.6, 6.2, 8.9, 0.7, 4.4, 6.7, 9.9, 0.7 State the median. Give your answer as a decimal.

- Find mean

Find the mean of two numbers: -4 and 5 (the first is negative four).

- Third tests

Third periodical tests are 98, 97, 86, 94, 90, 97, 91, and 94. Find the median of her grades and interpret the result.

- **75th percentile (quartile Q3)**



Find 75th percentile for 30,42,42,46,46,46,50,50,54

- **Median and modus**



Rachel rolled a die 50 times. The table shows the frequency of each face: Face Number: 1 2 3 4 5 6 Frequency: 8 7 5 11 6 13 Calculate the mode and median of the face numbers that Rachel rolled.

- **The data**



The data set represents the number of cars in a town given a speeding ticket daily for ten days. 2 4 5 5 7 7 8 8 8 12 What is the IQR?

- **Measuring data**



We gradually measured the values of 3,8,13,10,1,13,10,5,3,9. Calculate how much is the sum of the median and average of this file from the mode of this file.